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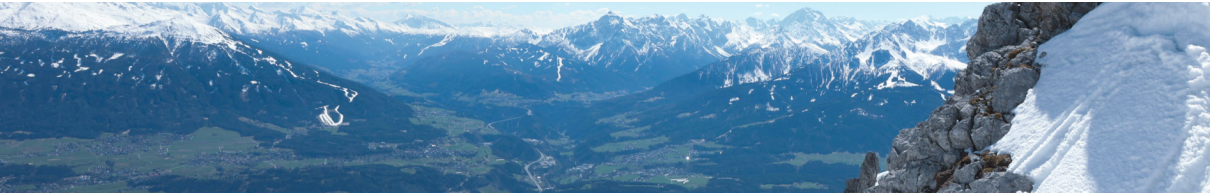


Rougelike, Working Title

VU Advanced C++ — Team 03

Martin Ehrle, Stefan Wagner, Robert Zacchia

Overview

- Gameplay concept
- Libraries and Frameworks
- Architecture
- ECS
- CI
- Work split
- Milestones

Gameplay Concept

- Rougelike 2D arena shooter
- Inspirations taken from Brotato[1]
- Waves of enemies
- Elite enemy every 5th wave
- Upgrade store after elite
- Scaling enemy strength and count
- Controls via mouse and keyboard

Brotato Example[2]

[1] https://brotato.wiki.spellsandguns.com/Brotato_Wiki
[2] https://store.steampowered.com/app/1942280/Brotato/

Gameplay Concept - Targets

- 3 different enemy types
- 2 different boss types
- 2 playable characters
  - melee
  - ranged
- Store upgrades
  - passive abilities
  - active abilities

Frameworks, Dependencies and other Ressources

Library	Purpose
SFML[1]	Multimedia Library
imgui[2]	Debug UI
Glaze[3]	JSON Library
Catch2[4]	Unit Testing
LibreSprite[5]	Pixel Art Tool
pre-commit[6]	Git Hooks (clang-format)
codecov[7]	Test Coverage

[1] https://www.sfml-dev.org
[2] https://github.com/ocornut/imgui
[3] https://github.com/stephenberry/glaze
[4] https://github.com/catchorg/Catch2
[5] https://libresprite.github.io/
[6] https://pre-commit.com
[7] https://about.codecov.io

Repository High-Level Structure

```
uibk_cpp_project/
├─ apps/
├─ assets/
└─ include/
   ├─ controller/
   ├─ game/
   └─ ui/
```

```
(continued)
│
├─ src/
│  ├─ controller/
│  ├─ game/
│  └─ ui/
└─ tests/
   ├─ controller/
   └─ game/
```

Entity Component System

- Entities
  - Entities are just IDs
  - *e.g. player, enemy, projectile, etc . . .*
- Components
  - Components are pure data
  - Properties an entity has
  - *e.g. component position with properties x and y*
- Systems
  - Systems perform operations on components of an entity
  - Contains all the logic
  - *e.g. system movement processes entities with position + velocity components*

Entity Component System - First Draft

Entities	Components	Systems
Player	Position	Input
Enemy	Velocity	AI
Projectile	Sprite	Movement
Weapon	Health	Collision
Item	Damage	Damage
...	Collision	Lifetime
	Lifetime	Pickup
	Weapon	Experience
	Stats	Render
	...	...

Entity Component System - Player

Entities	Components	Systems
Player	Position	Input
Enemy	Velocity	AI
Projectile	Sprite	Movement
Weapon	Health	Collision
Item	Damage	Damage
...	Collision	Lifetime
	Lifetime	Pickup
	Weapon	Experience
	Stats	Render
	...	...

Entity Component System - Enemy

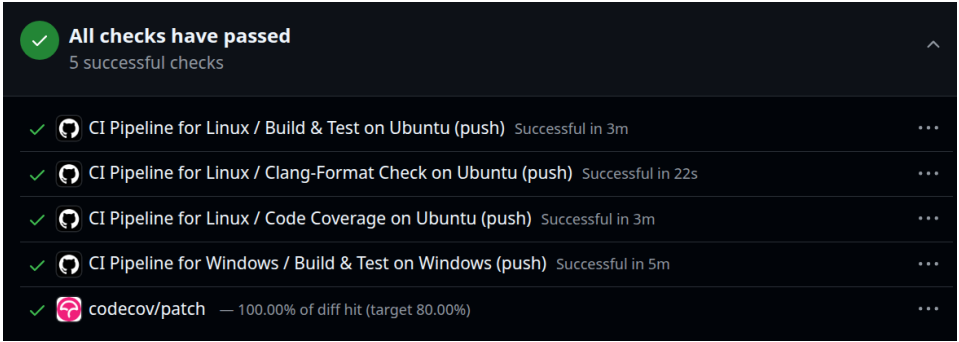
Entities	Components	Systems
Player	Position	Input
Enemy	Velocity	AI
Projectile	Sprite	Movement
Weapon	Health	Collision
Item	Damage	Damage
...	Collision	Lifetime
	Lifetime	Pickup
	Weapon	Experience
	Stats	Render
	...	...

Entity Component System - Projectile

Entities	Components	Systems
Player	Position	Input
Enemy	Velocity	AI
Projectile	Sprite	Movement
Weapon	Health	Collision
Item	Damage	Damage
...	Collision	Lifetime
	Lifetime	Pickup
	Weapon	Experience
	Stats	Render
	...	...

CI

- CI covers building, testing, test coverage, clang-format check
- Configured to run on Ubuntu and Windows container

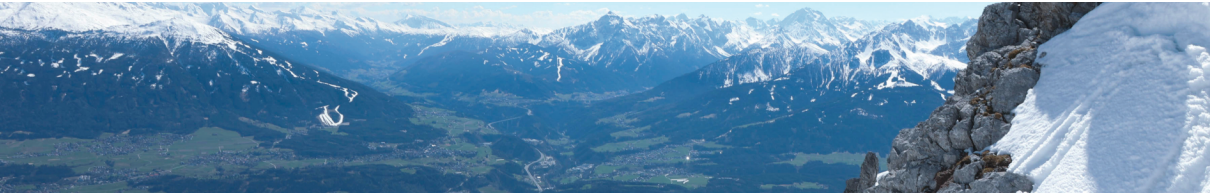


Ownership

Martin	Stefan	Robert
main menu	pause menu	game over screen
player controls	wave spawning	hitboxes & damage systems
enemy behaviour	progression store	progression
special ability A	special ability B	stage design
		special ability C

Milestones

- MS01 Shared Setup
  - minimal playable loop (character moves in empty space)
  - menu switching
- MS02 Gameplay
  - basic combat loop works
  - enemy spawning and attacking
  - transition between waves
  - menus are hooked up
- MS03 Finish
  - refinement
  - adding abilities
  - balancing



Thank you for your attention!

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